Solar Power, Explained for Everyone

Why Team Solar Foresight guesses tomorrow's sunshine power. Project: "Tomorrow's Sun, Today's Plan".

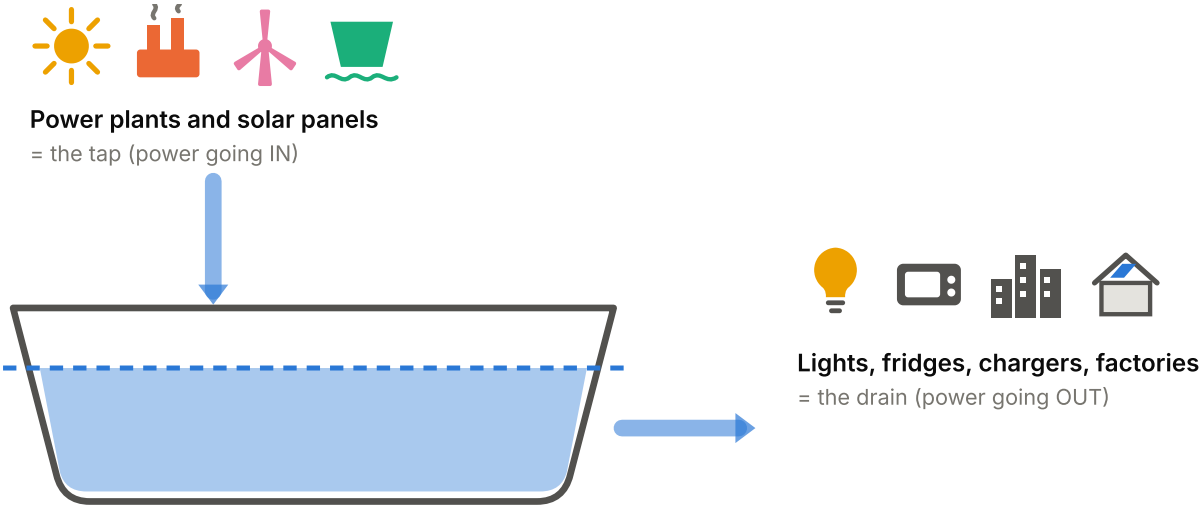
Team Solar Foresight · BUS 9430 · November 2026

How to read the tags. Every number says where it came from. Take your time. Read 1 section a day if you like.

- ✔ **SOURCE** a published source
- ● **OUR DATA** our own data pulls
- = **CALCULATED** simple math on sourced numbers
- ≈ **ROUGH ESTIMATE** a rough estimate or educated guess
- ? **NOT CONFIRMED** we could not confirm it, so treat it with care

1 The 1-minute story

Electricity is like water in a bathtub.



The grid is a bathtub. Plants fill it. Everything we plug in drains it.

- Electricity is used at the moment it is made. Very little can be saved for later.
- The water level must stay flat every second.
- Solar is a tap that opens with the sun. Clouds squeeze it.

- Solar is big: **25%** of California's power in 2025. At noon on our 4 sample days, solar matched **78% to 95%** of what CAISO's area was using.
- The power plan is locked the day before: about **97%** of tomorrow's power is lined up by 1 p.m. the day before. So we must guess tomorrow's sun today.

✓ **SOURCE** 25%: California Energy Commission, 2025

≈ **ROUGH ESTIMATE** 78% to 95%: 4 sample days; CAISO's solar file reads 12% to 13% above the U.S. Energy Information Administration

≈ **ROUGH ESTIMATE** 97%: see section 7

Our business problem

Tomorrow's sun is the hardest part of tomorrow's power plan to know. The plan is locked at 10 a.m. the day before. A wrong guess costs money.

Our business question

Can we guess how much solar power California will make in each hour of tomorrow, using only what is known by 10 a.m. today, almost as well as the experts at CAISO (the California Independent System Operator, the grid referee)?

Our goal: our average miss is at most 1.25 times CAISO's average miss.

Who is CAISO, and why does it exist?

CAISO is short for the California Independent System Operator. Think of an airport. Many airlines own the planes (the power plants). Other companies own the runways (the wires). If everyone moved on their own, planes would crash. So 1 neutral air traffic controller tells everyone when and where to go. For California's power, that is CAISO.

- Many companies make power. Many companies own the wires. Power cannot be stored in bulk, so supply must match use every second.
- CAISO runs the market where power is offered and bought, and keeps the balance. It is a not-for-profit, independent of the companies that make and sell power.
- Created by California's legislature in 1998, when the state opened electricity markets to competition. The Governor appoints its leaders.
- Looks after about **80%** of California's power.

✓ **SOURCE** Wikipedia, California Independent System Operator

2 The ruler: how we count electricity

Power is how fast. Energy is how much.

Think of a car. Speed is miles per hour. Distance is miles. **Power** is the speed of using electricity. **Energy** is the distance: how much was used over time.



1 watt (W)
the small tap

uses 1 Wh every hour

A bright LED bulb is 10 W: 10 Wh every hour



1 kilowatt (kW)
= 1,000 W

uses 1 kWh every hour

A microwave is about 1 kW. A typical home rooftop system is 7.7 kW



1 megawatt (MW)
= 1,000 kW

makes 1 MWh every hour

1 MW of solar powers about 173 U.S. homes (national average)



1 gigawatt (GW)
= 1,000 MW

makes 1 GWh every hour

California uses about 31 GW on average, 24 hours a day

Each step is 1,000 times bigger. Wh = watt-hour = 1 watt used for 1 hour. kWh = 1,000 Wh. MWh = 1,000 kWh. GWh = 1,000 MWh.

Energy over a longer time

1 TWh (terawatt-hour) = 1,000,000 MWh. We use TWh for a whole year.

California uses **273 TWh per year**. That is the same as **31 GW for 24 hours a day, 365 days a year** (31 GW x 8,760 hours = 273 TWh).

✔ **SOURCE** California Energy Commission, 2025 planning figure: 272,743 GWh

= **CALCULATED** 272.7 TWh ÷ 8,760 hours = 31.1 GW

Our rule: 1 unit per job

- An hour or a moment: **MW**. Whole state: **GW**.
- A day or month: **GWh**. A year: **TWh**.
- Homes: **kW** and **kWh**.
- We never mix GWh and TWh in 1 table.
- 1 MW for 1 hour = 1 MWh. So hourly MWh and MW are the same number.

✔ **SOURCE** 1 MW of solar powers about 173 U.S. homes: Solar Energy Industries Association, national average through Q2 2022

What each of our sources uses, and what we show

Where the number comes from	What it reports	Unit we show
Hourly solar count, U.S. Energy Information Administration	MWh for each hour (= average MW in that hour)	**MW**
CAISO market files	MW for each hour	**MW**
California Energy Commission yearly tables	GWh per year	**TWh per year**
Plant list, U.S. Energy Information Administration	Capacity in MW: the most a plant can make	**MW**
Weather forecasts, Open-Meteo	Sunshine strength in watts per square meter (W/m^2). 1,000 = full midday sun. Cloud cover in %.	**W/m^2, %**

Where the number comes from	What it reports	Unit we show
CAISO turned-down solar files	MW for each 5 minutes	**MWh** (MW ÷ 12)

Names to know. CAISO is short for the California Independent System Operator. Some data files, including ours from the U.S. Energy Information Administration, write **"CISO"**. That is the 4-letter code for the same grid area. Whenever you see CISO, read CAISO.

✓ **SOURCE** U.S. Energy Information Administration, hourly grid data guide

3 How much power does California use?

A lot, and it changes every hour.

273 TWh
per year

747 GWh
per day, on average

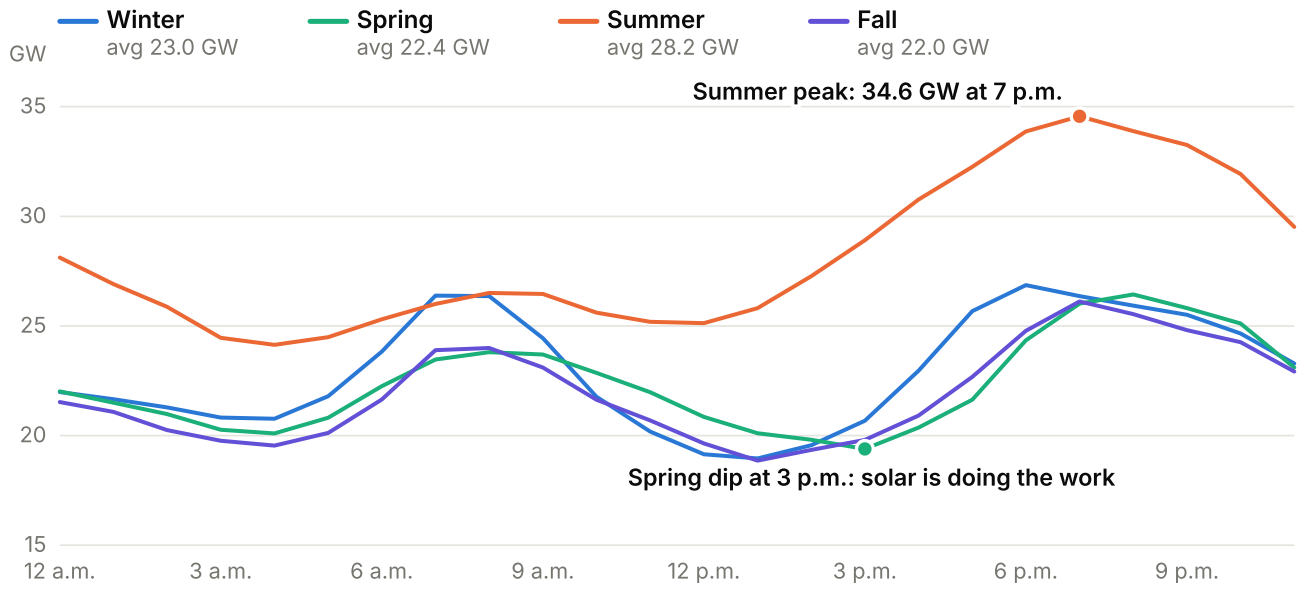
31 GW
every hour, on average

48.3 GW
biggest moment of 2024, 4 to 5 p.m. on Sep 5

✓ **SOURCE** California Energy Commission, 2025

= **CALCULATED** 273 TWh ÷ 365 days; ÷ 8,760 hours

✓ **SOURCE** CAISO 2024 market report: peak 48,323 MW



CAISO's area on 1 sample day per season (the 15th of Jan, Apr, Jul, Oct 2025). Sample days, not averages. ✔ **SOURCE** CAISO public daily files ≈ **ROUGH ESTIMATE** 1 day per season

- Lowest: early morning, and early afternoon when rooftop solar is working.
- Highest: 6 to 8 p.m. when people get home.

4 Where does it come from?

A mix of sources. Solar fills the middle of the day.

California's 2025 electricity mix: 267.7 TWh

Natural gas

81.3 TWh 30%

Solar (big farms)

66.5 TWh 25%

Wind

33.2 TWh 12%

Hydro (big dams)

29.9 TWh 11%

Nuclear



26.4 TWh 10%

Other (biomass, small hydro, unspecified)



17.3 TWh 6%

Geothermal

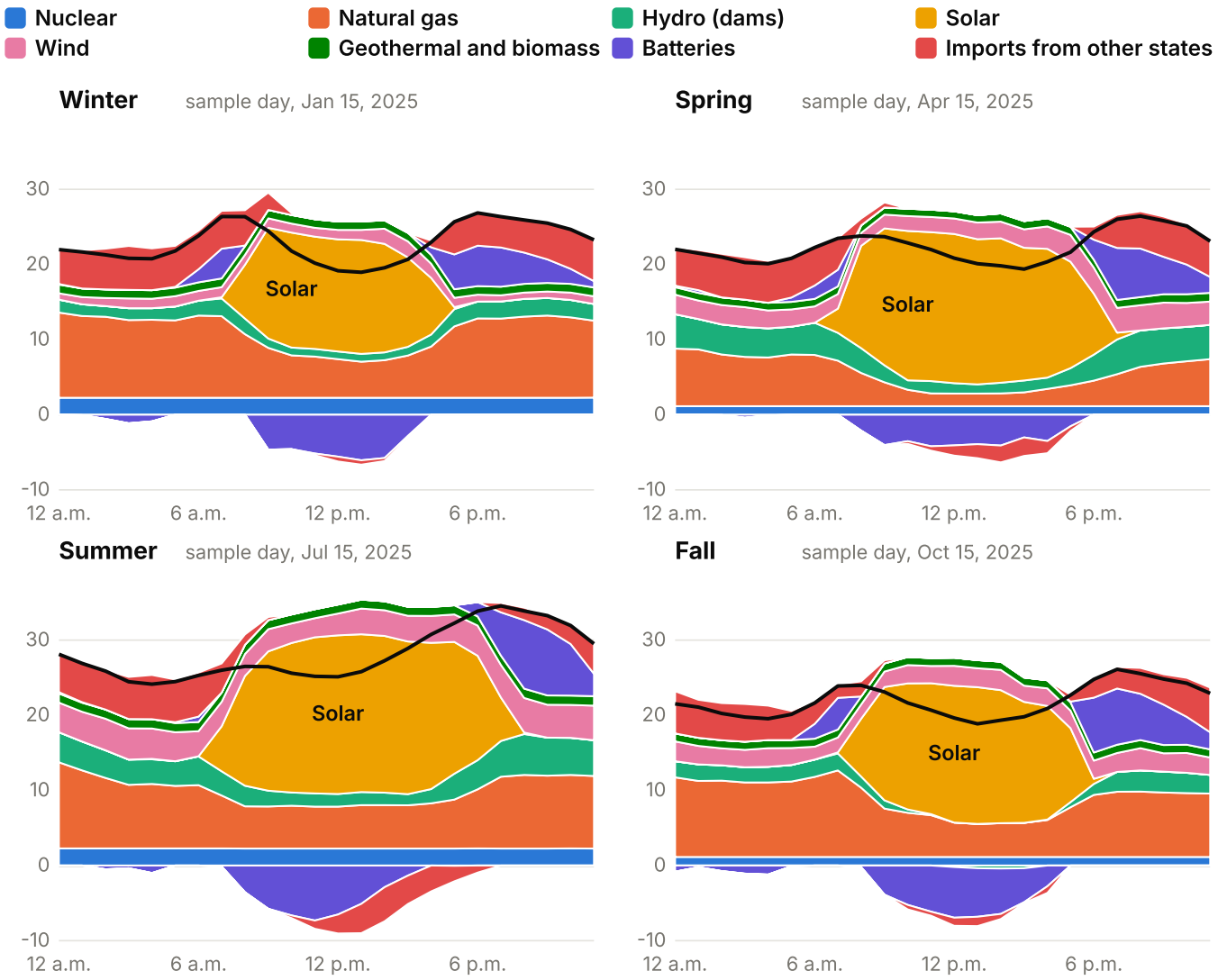


13.2 TWh 5%

✓ **SOURCE** California Energy Commission, 2025 Total System Electric Generation. Includes imports.

- About 23% (62 TWh) came from other states.
- Rooftop solar is not in these bars. It adds about 30 to 34 TWh more.

✓ **SOURCE** Rooftop estimates: California Energy Commission 30.3 TWh; U.S. Energy Information Administration 34.4 TWh



Layers = kinds of power, in GW. The black line = demand. Below the 0 line, batteries are filling up and power is leaving for other states. 1 sample day per season. ✔ **SOURCE** CAISO public daily files

≈ **ROUGH ESTIMATE** 1 day per season

- **Solar** fills the middle of the day. On the July sample day it made 220 GWh, 33% of that day's use. On the January day: 116 GWh, 21%.
- **Batteries** fill up at noon and give back from 6 to 9 p.m.
- **Natural gas** covers the evening and night.

5 Why does solar need a forecast?

The sun cannot be ordered around.

Solar is big and growing

- 25% of California's 2025 mix: the 2nd biggest source, after natural gas (30%).
- Solar capacity grew **24%** from Mar 2024 to Sep 2026: 21.0 GW to 26.0 GW.

✔ **SOURCE** California Energy Commission, 2025

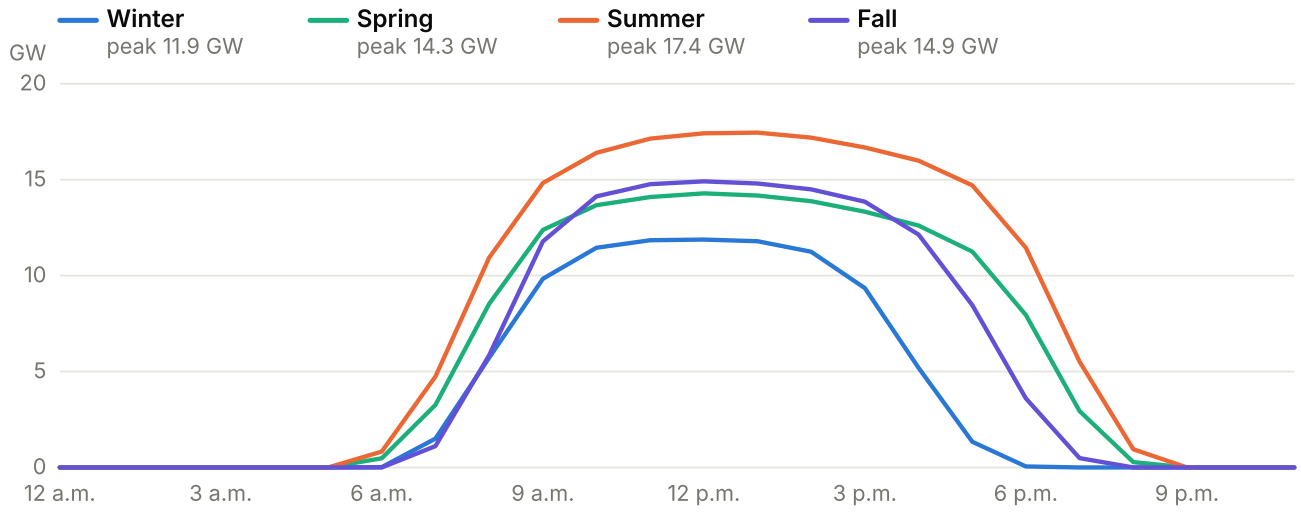
● **OUR DATA** monthly plant list from the U.S. Energy Information Administration, in our data

But it changes fast

- A gas plant takes hours to warm up, like an oven.
- So planners choose the day before which plants to start.
- On Apr 15, 2025 solar fell from **14.1 GW at 5 p.m.** to **0.9 GW at 7 p.m.**: 13.3 GW gone in 2 hours.

✔ **SOURCE** CAISO public daily file, Apr 15, 2025

= **CALCULATED** 14,148 MW minus 890 MW



Average solar output for each hour, over all 954 days in our data (Mar 8, 2024 to Oct 17, 2026), by season. Clock times are Pacific.

● **OUR DATA** our hourly solar table (from the U.S. Energy Information Administration)

- Same sun, different seasons. Summer peaks near **17.4 GW**. Winter peaks near **11.9 GW** and the sun is gone by about 4 p.m.
- That is why 1 forecast must learn all 4 seasons.

= **CALCULATED** average of every day in each season, from our table of 22,896 hours

6 Who is who

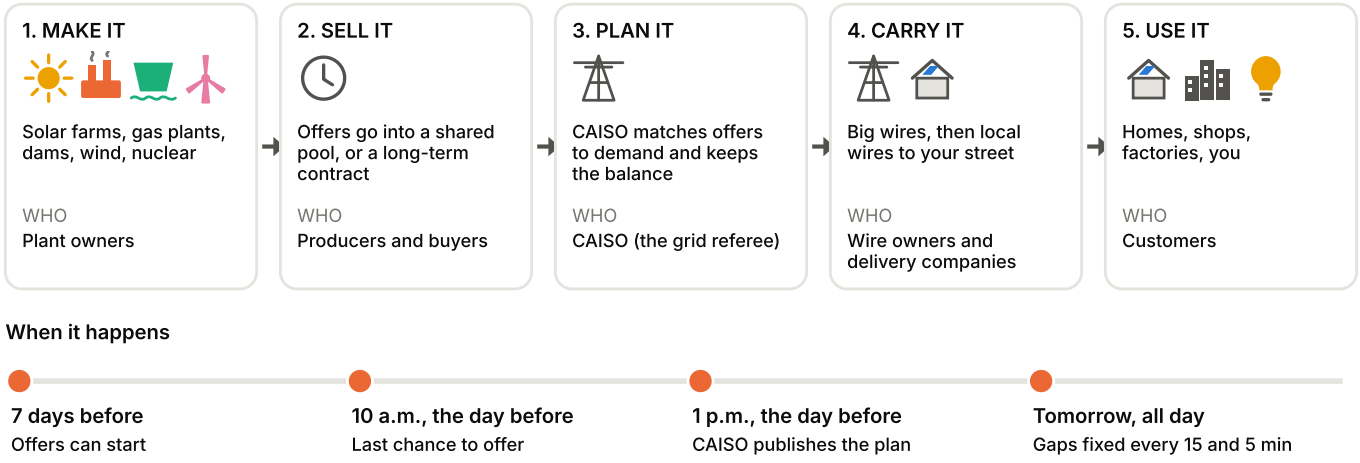
Everyone in the story, and their New York twins. (CAISO's own story is in section 1.)

✓ **SOURCE** New York Independent System Operator, 2019 blog post

✓ **SOURCE** Con Edison, Apr 2023: 55,000+ customer solar systems, 500+ MW

✓ **SOURCE** CAISO news release, May 1, 2026

✓ **SOURCE** California Public Utilities Commission, 2025 renewables report



From plant to plug, with the day-before timeline.

Job	California	New York
Grid referee. Keeps power in balance. Runs the markets.	**CAISO**, the California Independent System Operator. Not-for-profit.	**NYISO**, the New York Independent System Operator. Not-for-profit.

Job	California	New York
Delivery companies. Own the local wires and bills.	Pacific Gas and Electric, Southern California Edison, San Diego Gas and Electric. City utilities like Los Angeles Department of Water and Power, Sacramento Municipal Utility District.	**Consolidated Edison (ConEd)**: 3.5 million customers in New York City and Westchester. Public power: Long Island Power Authority, New York Power Authority.
Buying groups.	Community choice aggregators: city or county groups that buy power for residents, for example Clean Power Alliance of Southern California and MCE. They served 38% of load in 2025. Electric service providers: private companies selling to businesses.	Not covered here. Our project does not need it.
Producers. Own the plants.	Examples: Pacific Gas and Electric (Diablo Canyon nuclear plant), NextEra Energy (Desert Sunlight solar farms), Delta Energy Center (gas), California Department of Water Resources (dams), CalEnergy (geothermal), City of Santa Clara (gas).	Among the biggest: Constellation (Nine Mile Point, Fitzpatrick and Ginna nuclear plants), New York Power Authority (Niagara dams), Cricket Valley Energy (gas), CPV Valley (gas), National Grid Generation (gas), Astoria Energy (gas).
Rule makers and counters.	California Public Utilities Commission (rules for the big delivery companies). California Energy Commission (plans and counts). Federal Energy Regulatory Commission (oversees CAISO).	New York State Public Service Commission (oversees NYISO and power in the state). Federal Energy Regulatory Commission (oversees NYISO).
Number counters (federal).	U.S. Energy Information Administration: the federal government's energy statistics office. It counts and publishes hourly power for the whole country.	

✓ **SOURCE** Plant owners: gridinfo.com directory built on U.S. Energy Information Administration data (California rows are examples, not a ranking; New York rows are sorted by yearly output)

✓ **SOURCE** California Public Utilities Commission, 2025 report on investor-owned utilities: community choice groups served 38% of load in 2025

✓ **SOURCE** NYISO, Introduction to the NYISO: regulators and public power

✓ **SOURCE** CPUC and CEC power content labels name Clean Power Alliance and MCE

7 How power gets bought

A shared pool, not a 1-to-1 store.

Is it "buy and sell"? Yes, with 3 twists.

1. **It is a pool.** Producers put power into a shared pool. Buyers take from it. Nobody learns whose power they got.
2. **The buyers are delivery companies and buying groups.** Their job is to supply the homes on their wires.
3. **Most of the buying is done long before the day-ahead market.** The market fills the gaps.

Who buys, 2024 (share of power sold to customers)



- Pacific Gas and Electric, Southern California Edison, San Diego Gas and Electric: **47%**
- Community choice aggregators (city and county buying groups): **36%**
- Electric service providers (private sellers to businesses): **16%**
- Others: **1%**

✓ **SOURCE** California Public Utilities Commission, 2025 renewables report, p. 14

4 ways a delivery company gets power

A. Plants it owns

Dams and a nuclear plant owned by Pacific Gas and Electric are examples.

about 30%

B. Long-term contracts

A producer promises **ALL the power** of 1 plant to 1 buyer, at a fixed price per MWh, for years. Some deals cover only a part, and the producer sells the rest.

about 57%

C. Day-ahead market

Buy what is still missing, for tomorrow. Offers close at 10 a.m. the day before.

about 10%

D. Real-time market

Buy or sell the small gap between the plan and what really happens tomorrow.

about 3%

California, our rough estimate



- A. Plants the delivery company owns: **about 30%**
- B. Long-term contracts: **about 57%**
- C. Day-ahead market: **about 10%**
- D. Real-time market: **about 3%**

≈ **ROUGH ESTIMATE** our estimate, not a published number. Weakest guess: C (could be 5% to 20%)

- **A (about 30%).** Utility-owned plants made 75.9 TWh in California in 2024. Total supply, including imports, was 278.3 TWh. 75.9 ÷ 278.3 = 27%, rounded to about 30%.
  ✓ **SOURCE** EIA state electricity page, 2024
  ✓ **SOURCE** California Energy Commission, 2024 total system generation
  = **CALCULATED** 75.9 ÷ 278.3 = 27%
- **B (about 57%).** What is left after A, C and D: 100% minus 30% minus 10% minus 3%.
- **C (about 10%).** Power bought with no named plant behind it ("unspecified") was at least 4,051 GWh in 2024, which is 1.5% of supply. The day-ahead share is higher than that floor, because producers without a long-term deal sell there. Our guess is about 10%.
  ✓ **SOURCE** California Energy Commission, 2024
  = **CALCULATED** 4,051 ÷ 278,338 = 1.5%
  ≈ **ROUGH ESTIMATE** 10% is our judgment
- **D (about 3%).** The real-time share is the size of the day-ahead plan's miss: CAISO's plan was off by 2.3% to 3.4% on 4 sample days. ≈ **ROUGH ESTIMATE** average 3.0%, 4 sample days

California per year, in watts and in dollars

The same 4 ways, turned into energy, average power and money. **Dollars here are the value of the energy at $41 per MWh**, the 2024 average day-ahead price. That shows the size of what is at stake. It is not what anyone paid: owned plants and contracts have their own prices.

Way	Share	Energy per year	Average power, all hours	Value at $41 per MWh
A. Plants it owns	about 30%	82 TWh	9.3 GW	$3.4 billion
B. Long-term contracts	about 57%	156 TWh	17.8 GW	$6.4 billion
C. Day-ahead market	about 10%	27 TWh	3.1 GW	$1.1 billion
D. Real-time market	about 3%	8 TWh	0.9 GW	$0.3 billion
All California	100%	**273 TWh**	**31.2 GW**	**$11.2 billion**

= **CALCULATED** 273 TWh x share; TWh ÷ 8.76 = average GW; TWh x $41 per MWh

✓ **SOURCE** California Energy Commission, 2025 (273 TWh); CAISO 2024 market report ($41)

≈ **ROUGH ESTIMATE** the shares come from our rough estimate above

How big is a solar miss next to that?

If the solar guess is	Share of all California power	Energy per year	Average power	Value at $41 per MWh
Solar guess is the free copy of 2 days ago (6.1% miss)	**2.4% of all power**	**6.5 TWh**	**1.44 GW in a daylight hour**	**$265 million**
Solar guess is CAISO's, as published (4.7% miss)	**1.8% of all power**	**5.0 TWh**	**1.11 GW in a daylight hour**	**$204 million**
Solar guess is CAISO's, adjusted for turned-down solar (2.7% miss)	**1.0% of all power**	**2.9 TWh**	**0.64 GW in a daylight hour**	**$117 million**

= **CALCULATED** miss % x 23.6 GW average solar capacity x 4,489 daylight hours per year; 1 point of miss = 1.06 TWh

● **OUR DATA** miss rates are our corrected scores (section 10); capacity and daylight hours are from our hourly table

- **Every 1 point of miss is about 1.1 TWh a year.** That is about 0.4% of all California power, or roughly 236 MW in a daylight hour, worth about $43 million at $41 per MWh.

- **Where the miss lands.** Ways A and B (about 87%) are settled long before tomorrow. A solar miss lands on ways C and D, which are 35 TWh a year (13% of the total). CAISO's 5.0 TWh solar miss is about 14% of that.

  ≈ **ROUGH ESTIMATE** the two errors overlap and partly cancel, so read this as a size comparison, not a share

- **The real cost is smaller than the $41 and harder to pin down.** In 2024 prices were close across the markets: about $41 day-ahead, $40 in the 15-minute market and $39 in the 5-minute market. So a miss is not "buy at a high price". The cost is the extra work: fast plants kept ready, reserves, and imbalance. CAISO's real-time imbalance offset costs were **$234 million in 2024 and $205 million in 2025**. Bid cost recovery for commitment was $141 million and $137 million. Solar misses are one cause among several, and CAISO does not publish the split.

  ✓ **SOURCE** CAISO Department of Market Monitoring, 2024 and 2025 annual reports

  ? **NOT CONFIRMED** price averages are for the western market (WEIM) as a whole, and we did not confirm the CAISO-only 5-minute average

- **What our model could be worth.** CAISO already has its own forecast. A company or developer without one could copy 2 days ago (6.1% miss). A model at CAISO's level (4.7%) closes 1.4 points: about 1.5 TWh and $61 million a year at the average price. That is why an open model matters. ≈ **ROUGH ESTIMATE** a size, not a savings promise: our model has not been built or scored yet

- **Turned-down solar is a separate bill.** In 2025 about 3.5 TWh of solar was turned down by our count, about $143 million at $41 per MWh. That is an upper limit, because noon prices are lower. Most of it comes from too much solar for the demand and the wires, not from forecast errors. A better forecast helps at the edges.

  ● **OUR DATA** CAISO curtailment reports, solar only, 5 clock-change days left out

New York, for comparison



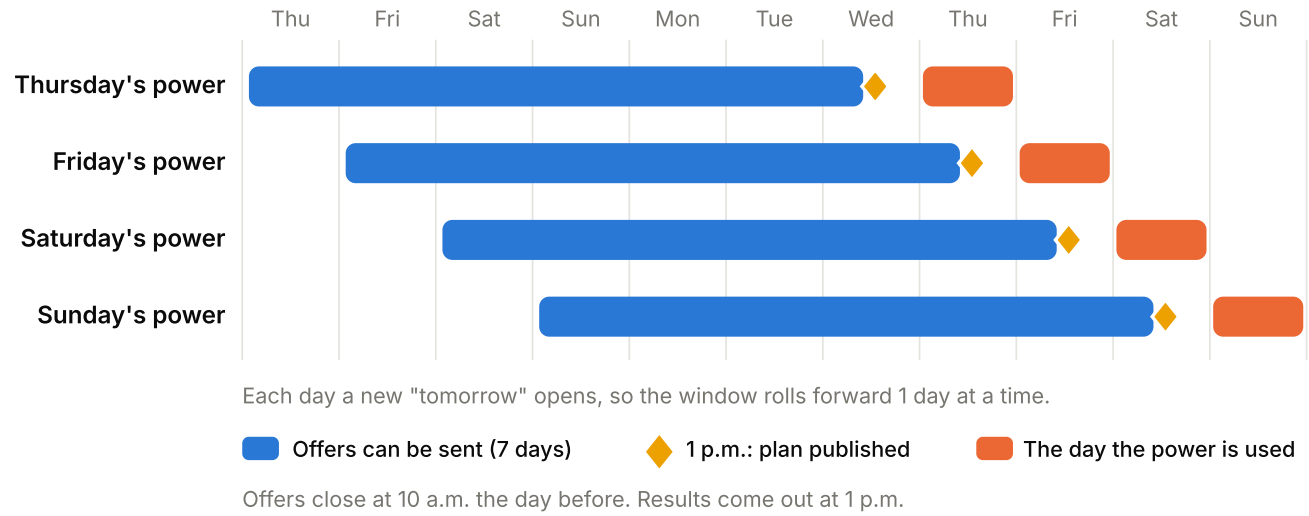
- Bilateral contracts: deals made outside the market, including long-term contracts: **about 45%**
- Day-ahead market: **about 51%**
- Real-time market (under 5%): **about 4%**

✓ **SOURCE** Independent Power Producers of New York, energy pricing paper (older, date not shown)

? **NOT CONFIRMED** we found no newer figure, so treat the New York split as old

How to read this. About 87% of tomorrow's power is settled long before the market opens, by own plants and long-term contracts. But every solar farm still makes whatever the sun gives, however it was bought. So someone must guess solar for the whole grid. That guess is the guess we test against CAISO's own day-ahead solar forecast.

The clock: a rolling week

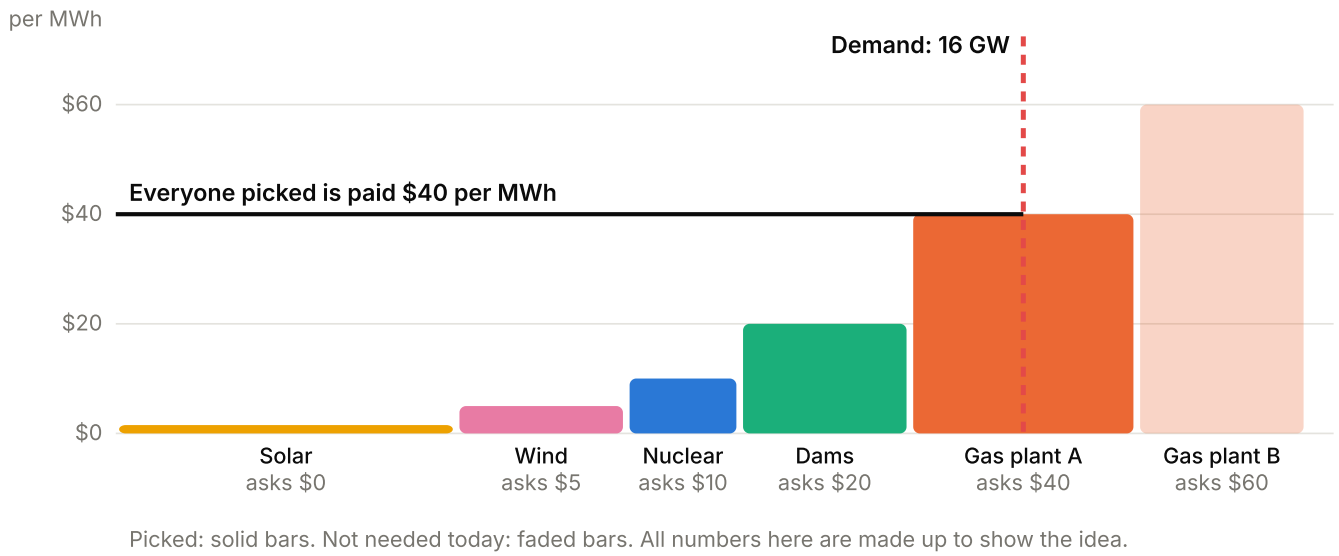


Each delivery day has its own window.

✓ **SOURCE** CAISO market rules, sections 30.1.1 and 30.5.1, and the CAISO Business Practice Manual for Market Instruments: offers up to 7 days ahead, close 10 a.m. the day before

- **How far ahead?** Up to **7 days**. A seller can send an offer for next Thursday's power starting last Thursday.
- **Rolling.** Every day, 1 window closes and a new window opens. So at any moment, 7 delivery days are open for offers.
- **Close: 10 a.m. the day before** (Pacific clock). Between 10 a.m. and 1 p.m. no new offers are taken.
- **1 p.m.:** CAISO publishes the plan for every hour of tomorrow. This is the **day-ahead market**.
  ✓ **SOURCE** CAISO market processes
- **Can offers change as the forecast changes?** Yes, until the 10 a.m. close a seller can send a new offer. After the plan is published, that day-ahead deal is set.
  ≈ **ROUGH ESTIMATE** educated guess: the CAISO rules we read do not say if a seller may edit an offer before the close
- **If an offer is not picked,** it is no longer binding. The seller can offer again in the real-time market, at a different price. ✓ **SOURCE** CAISO market rules, section 30.5.1
- **Can a buyer give some back?** Not by cancelling. If a buyer ends up using less than its plan, the extra is sold back in the real-time market at the real-time price, which can be higher or lower. If it uses more, it buys the rest the same way. ✓ **SOURCE** CAISO real-time load settlement paper

Who wins the match: the lowest price



Picked: solid bars. Not needed today: faded bars. All numbers here are made up to show the idea.

How offers are picked. All numbers are made up. ≈ **ROUGH ESTIMATE** made-up prices

- CAISO lines up all offers from cheapest to most expensive. It picks the cheapest first, until there is enough power for everyone.
- Everyone picked is paid **the same price**: the price of the last offer that was needed.
- Solar and wind ask close to $0 per MWh because the sun and wind are free. So they go first and pull the price down. That is why a sunny afternoon often has cheap power.
- In 2024, CAISO's day-ahead price averaged **$41 per MWh**. CAISO prices also change by place; we ignore that here.

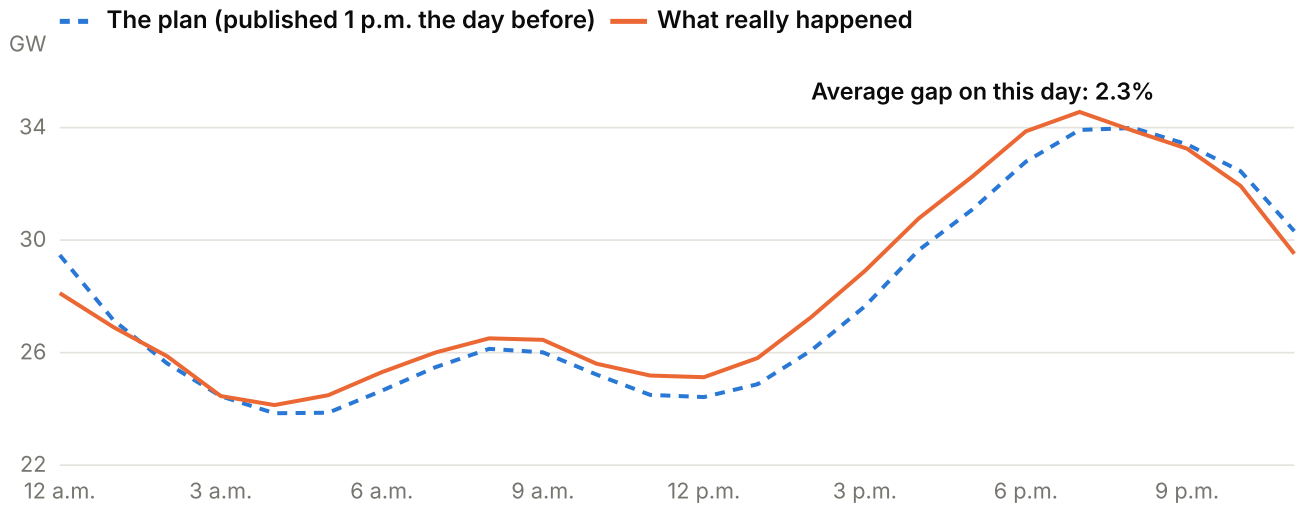
✓ **SOURCE** ISO New England, "How resources are selected and prices are set" (another grid operator, same idea)

✓ **SOURCE** CAISO 2024 market report: $41 per MWh

Tomorrow: the real-time market and the "plant reset"

- Real use never matches the plan exactly. Example (made-up numbers): the plan said 26,000 MW at 7 p.m. Real use was 26,800 MW. The 800 MW gap (3%) is bought in the real-time market.
- **What the reset is.** Every 15 minutes and every 5 minutes, CAISO's computer checks real use and sends each plant a new target: "make a bit more" or "make a bit less". It is like a gas pedal that holds a steady speed.
- **Why.** To keep supply equal to use every second, at the lowest cost. Plants that can move fast do the work: batteries, dams, some gas plants. Big gas plants cannot start fast. That is why the plan is made the day before. Solar can be turned down fast. That is curtailment (section 8).

✓ **SOURCE** CAISO market processes: real-time market dispatches every 15 and 5 minutes



The plan (dashed) and what happened (solid), July 15, 2025, CAISO's area.

✔ **SOURCE** CAISO public daily file

about 97%

about 97% planned the day before

about 3% fixed during the day

≈ **ROUGH ESTIMATE** CAISO's day-ahead demand plan was off by 2.3% to 3.4% on 4 sample days (average 3.0%)

✔ **SOURCE** CAISO paper on real-time load settlement: 0.8% in 1 worked example

8 When the guess is wrong

Too high means last-minute buying. Too low means extra power nobody planned for.

Each pizza = 10 slices. Think of slices as units of solar power.

Guess too HIGH: we planned 100 slices (10 pizzas)



Only 70 slices show up.
Buy the 30 missing slices at the last minute. Costs more, often dirtier.

Guess too LOW: we planned 70 slices (7 pizzas)



100 slices show up.
30 are not needed. They are sold, stored, sent away, or solar is turned

Either way, a better guess means fewer last-minute buys and less waste.

A missed solar guess in pizza terms.

Too low: is the extra solar lost?

> **Short answer: mostly no. It is sold and used. It is "lost" only at the very last step, and then the power was never made.** The sun makes what it makes. A low guess only means the day-ahead plan had less solar in it than showed up.

What happens to the surplus, in order:

1. **It is sold.** A farm that makes more than its day-ahead plan sells the extra in the real-time market at the real-time price. At noon that price can be very low, even below $0.
   ? **NOT CONFIRMED** this is the standard two-settlement rule; we did not check the exact CAISO tariff line
2. **Other plants back down.** Every 5 minutes CAISO sends each plant a new target. Gas plants and dams get "make a bit less". ✔ **SOURCE** CAISO market processes
3. **Batteries fill up.** Charging peaks around noon, when prices are lowest. Batteries in CAISO's area grew from 8.0 GW in 2023 to 11.6 GW in 2024.
   ✔ **SOURCE** U.S. Energy Information Administration, Today in Energy (curtailment)
   ✔ **SOURCE** GridStatus, spring 2026 report
   ? **NOT CONFIRMED** CAISO market monitors report about 25.7 GW at the end of 2025, but we could not confirm whether that covers the whole western market
4. **It is sent to neighbors.** Through the Western Energy Imbalance Market, CAISO sells extra midday power to other states. That avoided 274 GWh of turn-downs in 2024, about 8% of the total. In April 2026 the midday flow turned to net imports, so this outlet got smaller.
   ✔ **SOURCE** U.S. Energy Information Administration, Today in Energy (curtailment)
   ✔ **SOURCE** GridStatus, spring 2026 report
5. **Only then is solar turned down.** It happens through price signals (a farm's own bid says "stop below this price") or, rarely, a direct order. In 2025 about 89% of wind and solar turn-downs came from the farms' own bids.
   ✔ **SOURCE** U.S. Energy Information Administration, Today in Energy (curtailment)
   ✔ **SOURCE** CAISO Department of Market Monitoring, 2025 annual report

What "lost" really means, and why nothing needs to be thrown away

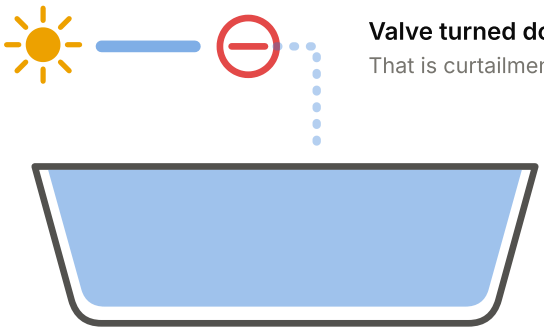
- **Nothing is dumped.** When a farm is told to make less, its inverters move the panels away from their best operating point, so the panels deliver less. The sunlight is simply not turned into power. There is nothing to get rid of and nothing to make safe.
  ✔ **SOURCE** arXiv paper on PV power curtailment (maximum power reference tracking)
- **What is lost** is the clean energy that could have been made, and the owner's sales for those hours. It is a money and climate loss, not a safety event.
- **Power cannot sit in the wires.** What goes in must come out at the same instant: as use, as battery charging, or as exports. That is why the answer to a surplus is to use it, store it, send it away, or not make it.

How the grid is protected from too much power

If more power goes in than comes out, the grid frequency rises above 60 Hz. CAISO says that "can lead to grid instability". The opposite case is worse: a long shortage can lead to cascading blackouts. ✔ **SOURCE** CAISO frequency response issue paper The protection comes in layers, slowest first:

- **1. The plan (day-ahead).** Sets most of tomorrow's mix. This is the layer our forecast helps. The better the plan, the less the later layers must do.
- **2. The market, every 5 and 15 minutes.** The steps above.
- **3. Regulation, every 4 seconds.** A control computer sends small up and down signals to plants, with the full response within about 10 minutes. ✔ **SOURCE** CAISO frequency response issue paper
- **4. Governors, within seconds, with no human.** Plants react by themselves to a frequency change as small as 36 mHz. CAISO relies on these settings.
  ✔ **SOURCE** CAISO frequency response issue paper
- **5. Turn-down orders.** Described above.
- **6. Last-resort protection.** Safety relays disconnect equipment if frequency leaves safe limits.
  ? **NOT CONFIRMED** general engineering practice; we did not check the exact settings for the Western grid
- **Rooftop solar** cannot be commanded one system at a time. It shows up as lower demand (section 9), so the grid handles it through the layers above.

Curtailment: solar turned down on purpose



Valve turned down on purpose
That is curtailment.

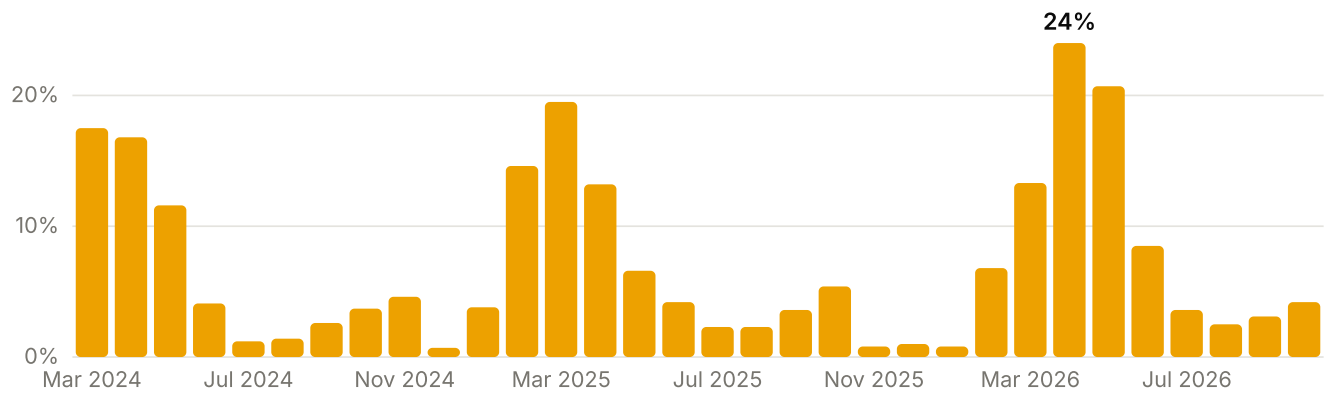
The tub is already full:
1. People are not using much power right now
2. Wires are full
3. Some gas plants must stay on for the evening

When the tub is already full, CAISO asks solar farms to produce less.

- In our period, **7.5%** of the solar that could have been made was turned down. In April 2026: **24%**.
- Most happens in spring: lots of sun, little demand.
- In 2024, 3.4 TWh of wind and solar was turned down in CAISO's area. 93% of it was solar.

● **OUR DATA** CAISO curtailment reports and our hourly table

✔ **SOURCE** U.S. Energy Information Administration, Today in Energy, 2025



Share of possible solar that was turned down, by month. Mar 2024 to Oct 2026.

9 Big solar farms vs rooftop solar

Same sun. Very different counting.

BIG SOLAR FARMS



Fields of panels. Power goes onto the wires.
CAISO can see and count them every hour.
Our "answer key" counts only these.

Makes about 25% of its full-sun maximum over a year (our data).
In full sun, 1 MW makes 1 MWh every hour.

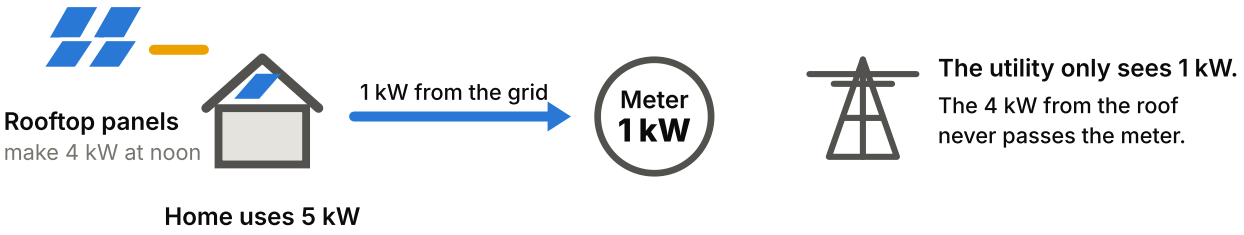
ROOFTOP SOLAR



Panels on homes and shops (each under 1 MW).
The home uses it first. Extra goes to the grid.
Nobody counts it on its own. The grid just sees less demand.

Makes about 17% of its full-sun maximum over a year (article estimate).
In full sun, a 7.7 kW home system makes about 7.7 kWh every hour.

Our forecast covers the big farms. Rooftop solar shows up as lower demand.



Made-up numbers. 5 kW used minus 4 kW from the roof = 1 kW from the grid. Rooftop looks like lower demand.

How rooftop solar hides inside the meter.

- **Rooftop = under 1 MW per system.** California had about **20 GW** of it in 2025. It made about **34 TWh**, 37% of all U.S. rooftop solar.
- About 2 million systems. About 1 in 6 California homes. Averaged over all systems (homes and businesses), about 9 kW each. A typical U.S. home system is smaller: 7.7 kW.
- Since April 2023 the credit for sending extra power out is lower, so more homes add batteries.
- **Why the grid cannot see it.** The utility's meter sits at the home's front door. It counts only what the home takes from the grid. So rooftop power looks like the home used less.
- **Why the official count leaves it out.** The U.S. Energy Information Administration counts plants of 1 MW or more in its main power numbers. Rooftop systems are under 1 MW, so they are estimated separately and usually left out of the main count.

✓ **SOURCE** U.S. Energy Information Administration: 20 GW, 34,411 GWh, 37%, under 1 MW

✓ **SOURCE** Wind-Works article: 18,000 MW, nearly 2 million systems, 1 in 6 homes, 17% capacity factor

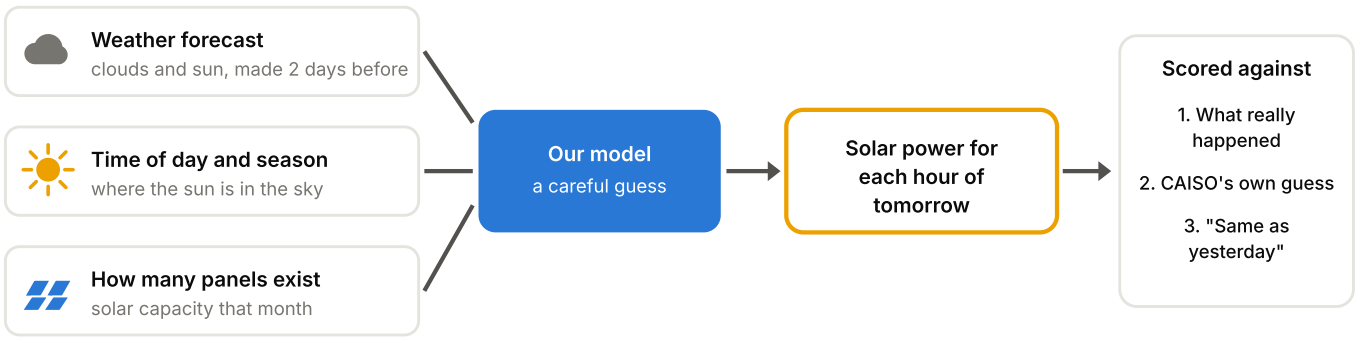
≈ **ROUGH ESTIMATE** 9 kW = 18,000 MW ÷ about 2 million systems

✓ **SOURCE** Lawrence Berkeley National Laboratory, Tracking the Sun: U.S. median home system 7.7 kW, 2025

? **NOT CONFIRMED** "1 in 6 homes" and "2 million systems" come from 1 article, not state data

10 Our project

A careful guess for each hour of tomorrow, and a fair scorecard.



What our forecast does.

Why it matters

- Fewer last-minute power buys.
- Less power wasted.
- Fewer backup gas plants running.

Who it helps

- Grid planners.
- Companies that buy power.
- Anyone who pays an electric bill.

How it is different

- Only free public data.
- A fair scorecard against CAISO and simple guesses.
- Every step shown, so anyone can check it.

Who else forecasts solar?

Who	What it is	How we differ
Solcast	A paid service. Solar forecasts from 5 minutes to 14 days ahead. Energy traders use it to place day-ahead bids.	We use only free public data and publish every step.
Open Climate Fix (Open Quartz)	A nonprofit with an open-source solar forecast. Global, 48 hours ahead. Trained on UK data and checked only in the UK.	Ours is built for the CAISO area and scored against CAISO's own day-ahead forecast.
CAISO's own forecast	The forecast CAISO publishes for tomorrow's solar.	This is the score we try to match.

✓ **SOURCE** Solcast website ✓ **SOURCE** Linux Foundation Energy article on Open Climate Fix

? **NOT CONFIRMED** vendor claims from their public pages. We did not test anyone's accuracy

Why not forecast demand instead?

Fair question. Here is the comparison on the 4 sample days we have (1 per season), next to our solar scores.

	Demand (total use)	Solar
Typical miss in MW	about 650 to 750 MW	about 1,100 MW in a daylight hour

	Demand (total use)	Solar
Miss as a share of what is forecast	2.3% to 3.4% of demand	about 10% of solar output (1.1 GW out of about 11 GW)
Fastest move in 2 hours	demand up 1.9 to 2.7 GW	solar down 7.2 to 9.2 GW
What drives it	temperature, habits, holidays	clouds, over the whole state at once

≈ **ROUGH ESTIMATE** demand column: CAISO day-ahead demand plan against actual, 4 sample days (the 15th of Jan, Apr, Jul, Oct 2025)

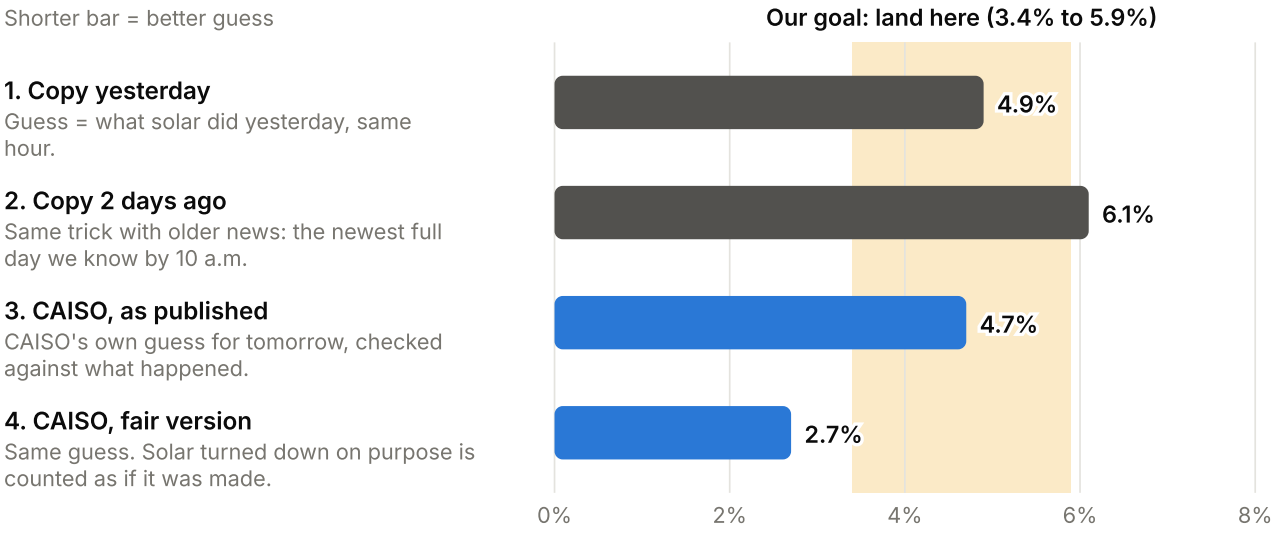
✓ **SOURCE** CAISO public daily files: demand, demand plan, solar

● **OUR DATA** solar column: CAISO as published, 4.7% x 23.6 GW; about 11 GW is our average daylight solar output

- **We are not saying demand is solved.** One outside test, July 4 to 12, 2024, found CAISO's own load forecast about 4.55% off on average and 3.2 GW too high at the July 4 peak. So demand misses are real, and larger on holidays and heat waves.
  ✓ **SOURCE** Yes Energy, CAISO demand forecast accuracy blog, 2024
  ? **NOT CONFIRMED** a company that sells a competing forecast wrote it, one week only
- **Solar is the bigger miss for its size.** About 10% of solar output, against 2% to 3% of demand. In MW the 2 are not far apart, but a solar miss lands on the part of the grid that moves fastest.
- **Solar moves 3 to 4 times faster.** A big gas plant needs hours to warm up. Demand rises 2 to 3 GW in 2 hours, which planners can follow. Solar can fall 7 to 9 GW in 2 hours at sunset. That drop has to be planned the day before.
- **The grid really runs on demand minus solar.** Solar is the part of that gap that swings the most, so a better solar guess shrinks the gap the other plants must cover.
- **Demand is a crowded field.** CAISO, utilities and paid vendors all publish demand forecasts. Free weather data feeds a solar forecast directly, so an open model can be tested fairly against CAISO.
- **A natural next step.** Our table already holds temperature forecasts, the main driver of demand. The same table could support a demand model later. That is outside this project.

The scorecard: 4 guesses, 1 ruler

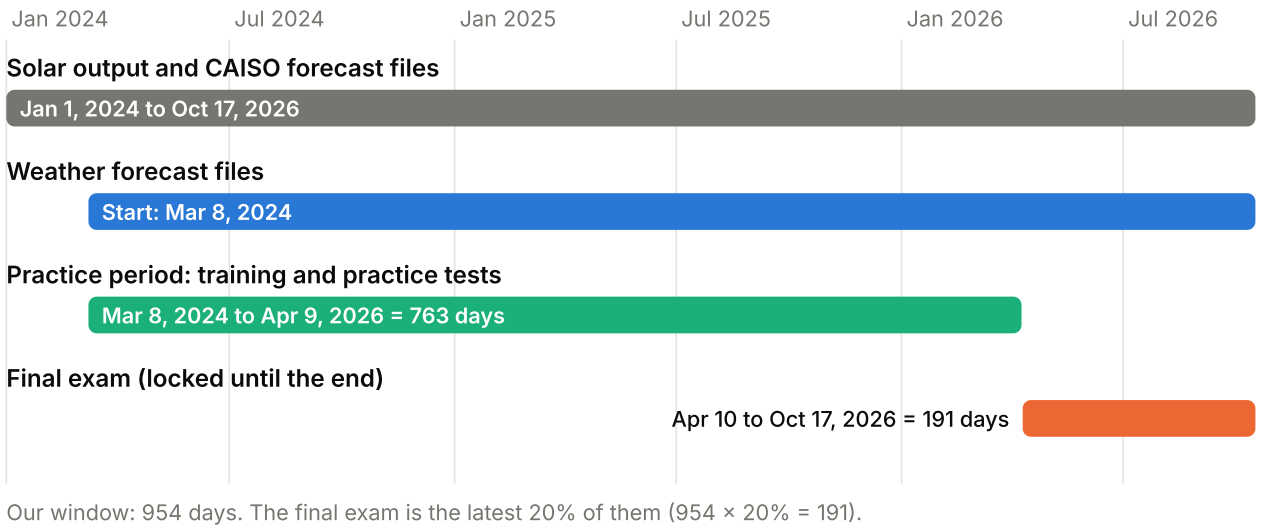
Each bar is 1 way to guess tomorrow's solar for each daylight hour. The bar shows how far off that guess was, on average. **Shorter is better.**



Average miss, in % of all solar plant capacity. Daylight hours, Mar 2024 to Apr 9, 2026 (9,083 hours over 754 days). ● **OUR DATA** our reference scores

- **What "miss" means.** The average gap between the guess and what happened, divided by all solar plant capacity. Example with made-up capacity: if plants can make 25,000 MW at most, a 5% miss is 1,250 MW off in an average daylight hour. = **CALCULATED** 5% x 25,000 MW = 1,250 MW
- **Bars 1 and 2** are free tricks anyone can do. Bar 2 exists because "yesterday" is not fully known at 10 a.m., so the fair copy is 2 days ago.
- **Bars 3 and 4** are the experts at CAISO. 2 versions, because solar is sometimes turned down on purpose (section 8). The meter then shows less than the sun could make. Bar 3 counts the meter as it is. Bar 4 adds the turned-down solar back, so CAISO looks better.
- **Our goal:** at most 1.25 times CAISO's miss. That is **3.4%** (1.25 × 2.7%) to **5.9%** (1.25 × 4.7%): the orange band. = **CALCULATED** 1.25 × 2.73 = 3.4; 1.25 × 4.68 = 5.9

Our time range, and why



Our window: 954 days. The final exam is the latest 20% of them (954 × 20% = 191).

Each bar shows when a kind of data is available.

- **Why start Mar 8, 2024?** The weather forecast archive is complete from that day. We need weather and solar data for the same hours.
- **Why end Oct 17, 2026?** We pulled the data on Oct 27 and 28, and the end date came from the pull date. From there we looked back as far as our sources allowed. The weather forecast archive is the most limited source: it starts Mar 8, 2024. The U.S. Energy Information Administration can revise its newest numbers, so the last few days may change a little.

  ● **OUR DATA** our data FAQ, Q4.4 ✔ **SOURCE** our data FAQ, Q6.5: values can be revised
- **Why is the final exam the latest 191 days?** In real life we only know the past. So the test is always the newest days, and we never train on them. 20% of 954 days = 191.
- **Why no winter in the exam?** The newest 191 days are April to October. Moving the exam to include winter would mean testing on old days and training on newer ones, which is not how a real forecast works.
- **What we do about it.** All 180 winter days (Dec to Feb) in our data sit in the 763 practice days, so winter is learned and tested there. If the 4 practice tests are the last 4 three-month blocks, 2 of them include winter: Oct 10, 2025 to Jan 8, 2026 (39 of 91 days) and Jan 9 to Apr 9, 2026 (51 of 91 days). We will say clearly in our results that winter was tested only in practice.

  = **CALCULATED** Dec, Jan, Feb days in the practice period = 180

  ? **NOT CONFIRMED** practice-test dates are not written down yet, so this is our assumption

● **OUR DATA** Our data FAQ, section 4

11 Our data

5 sources. Each source has a job.

	What it is	Where it comes from	What it is for	What else we could have used, and why not
	**1. The answer key** How much solar power was really made, each hour.	U.S. Energy Information Administration: the federal government's energy statistics office. Its hourly grid data. (Files say "CISO", which means CAISO.)	The number we forecast. Every guess is scored against it.	**Skipped:** CAISO's own count of actual solar. We keep it as a backup. It covers only 3 areas and shows peaks about 12% to 13% higher. The federal source has fixed definitions and public rules. ● **OUR DATA** Our data FAQ, Q3.2: we did not record whether the team compared the two
	**2. The scoreboard** CAISO's own guess for tomorrow's solar, made the day before.	CAISO, through its public data portal (OASIS).	The score to match. Never used to train our model.	**Skipped:** other forecasts. The business question is about beating CAISO's.
	**3. Weather forecasts** Cloud cover and sunshine strength as forecast 1 and 2 days ahead, at 51 spots in California.	Open-Meteo (free service), using the European weather model ECMWF.	The clues for our guess. We use the forecast made 2 days ahead, which is ready before the 10 a.m. deadline.	**Skipped:** measured sunshine from the National Renewable Energy Laboratory database. It records what happened, not what was forecast, so it would be too good to be true. Also skipped: other weather models (1 mixes models, 1 has no day-2 forecast, 1 returned no data).

	What it is	Where it comes from	What it is for	What else we could have used, and why not
	4. How many panels exist A monthly list of every power plant and its capacity.	The same U.S. Energy Information Administration that provides the answer key. Its monthly plant survey (form 860M).	Divides errors by capacity so months compare fairly. Places the weather spots near plants.	**Skipped:** CAISO's own capacity numbers (1 number a month in a PDF, no plant list). Used only to cross-check. The yearly version of the same survey is too slow: yearly, not monthly.
	5. Turned-down solar When solar was turned down on purpose, and by how much.	CAISO's curtailment workbooks (2024 to 2025) and daily renewable reports (June 2025 on).	Flags hours when output was lower than the sun allowed. We report results with and without them.	**Skipped:** CAISO's daily curtailment PDF. It returned no data for 2026.

OUR DATA Our data FAQ, sections 1 to 3

SOURCE Open-Meteo: free for non-commercial use, credit required (CC BY 4.0)

12 What could go wrong, in plain words

7 things that could trip us up, and what we do about each.

1. The weather forecast is wrong

WHY We rely on a free forecast made by someone else. A bad forecast means a bad solar guess.

WHAT WE DO We use the forecast made 2 days ahead, test the 1-day forecast as a comparison, and report errors by weather type. Forecast sunshine already tracks real output well: 0.88 on a scale where 1.0 is perfect.

2. We have little history

WHY The weather archive starts Mar 8, 2024. That is 954 days. Each season shows up only 2 to 3 times.

WHAT WE DO Keep the model simple. Test only in time order, never shuffled.

3. No winter in the final exam

WHY The exam is the newest 191 days, Apr 10 to Oct 17, 2026. A real forecast only knows the past, so the test must be the newest days. That leaves out winter, when solar is lowest and days are shortest.

WHAT WE DO All 180 winter days sit in the practice period, so winter is learned and tested there (in 2 of the 4 practice tests, if they are the last 4 three-month blocks). We report winter results separately and say clearly they come from practice, not the final exam.

4. Turned-down solar hides the truth

WHY The meter counts what was delivered, not what the sun allowed. Up to 24% in a spring month.

WHAT WE DO Flag those hours. Report results with and without them.

5. Clocks do not match

WHY Our main solar count (from the U.S. Energy Information Administration) ran 1 hour behind the sun and the other sources until Dec 2, 2025. Since then it is on time. A 1-hour slip ruins results.

WHAT WE DO We compared it hour by hour with CAISO's own count, and with the true time of solar noon. Before Dec 2, 2025 the files line up only when we move our count back 1 hour (0.998 match). After that date we do not move it (0.998 match; moving it would drop the match to 0.81). All sources then use 1 clock (UTC). We do not know why the count changed, and the team still confirms the rule.

6. Missing data

WHY 226 of 22,896 hours (0.99%) have a gap: 129 weather hours, 72 CAISO plan hours and 25 solar count hours. The 129 weather hours are in April 2026, inside the exam period.

WHAT WE DO We do not invent values. Short gaps are filled, long gaps are dropped, and every choice is written down.

7. Rooftop solar is invisible, and solar keeps growing

WHY Rooftop solar shows up as lower demand. Big-farm capacity grew 24% in 2.5 years.

WHAT WE DO Divide errors by capacity so months compare fairly. State clearly that rooftop is not in our numbers.

13 Quick answers

9 questions people ask.

▼ Why not just look at the sky?

1 sky covers a few miles. Solar farms are spread across a huge state. And the plan is made the day before, so we need to know today what tomorrow's sky will do. Weather forecast models do that.

▼ Why not use last year's weather?

The sun's path repeats every year. Clouds do not. A sunny Aug 3 last year says nothing about tomorrow. What repeats is the time of day and the season, and our model already knows those.

▼ Why not copy yesterday?

It works decently: an average miss of 4.9%. But it fails when the weather changes. It is the minimum score we must beat.

▼ Why not just use CAISO's forecast?

That is the score we compare ourselves against. Our job is to see whether free public data can get close.

▼ Why only solar, not total demand?

Demand is not solved, but solar is the bigger and faster miss. On our 4 sample days CAISO's demand plan missed by 2.3% to 3.4% (about 700 MW). Solar misses about 10% of its output (about 1,100 MW), and solar can fall 7 to 9 GW in 2 hours while demand moves 2 to 3 GW. Demand forecasting is also crowded: CAISO, utilities and vendors all do it. Free weather data drives solar directly, so an open model can be tested fairly. See section 10.

▼ If we guess too low, is the extra solar lost?

Mostly no. It is sold in the real-time market, other plants back down, batteries charge, and neighbors buy some. Only if none of that works is solar turned down, and then the power is never made: nothing is dumped. See section 8.

▼ What is a solar miss worth in dollars?

Every 1 point of miss is about 1.1 TWh a year, about $43 million at $41 per MWh. CAISO's 4.7% miss is about 5 TWh, about 1.8% of all California power. The true cost is lower than that and not published: CAISO's real-time imbalance offsets were $205 million in 2025. See section 7.

▼ Why California and not New York?

Short answer: California has the most solar, and its grid gives away the data we need for free. A bigger solar business means a better guess is worth more.

- **California is about 6 times bigger in solar.** California: more than **49,000 MW** of solar, rank 1 in the U.S. New York: **7,922 MW**, rank 7. 49,000 ÷ 7,922 = about 6.
- **Solar matters more there.** Solar makes more than **31%** of California's electricity. In New York it is **7.56%**. A miss in the solar guess hurts more where solar is bigger.
- **California is the bigger power market.** In 2024 California made **214.2 TWh** of electricity and New York made **129.0 TWh** (all sources).
- **Free data and a score to beat.** CAISO and the U.S. Energy Information Administration publish hourly numbers for free. CAISO also publishes its own day-ahead solar forecast. That is the score we try to match.

✓ **SOURCE** Solar Energy Industries Association, California (data through Q4 2024) and New York (data through Mar 2026) state pages

✓ **SOURCE** U.S. Energy Information Administration: 2024 net generation

= **CALCULATED** 49,000 ÷ 7,922 = about 6

? **NOT CONFIRMED** The team's original reason is not written in our project documents, so this is our best explanation. The two solar figures are from different dates. We did not check what free solar data New York's grid operator offers

▼ Why 10 a.m.?

Offers for tomorrow close at 10 a.m. Pacific the day before. Anything we use has to be known by then, or our results would look better than real life. CAISO publishes the plan at 1 p.m.

Sources

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37. Our project's data FAQ (docs/DATA_FAQ.md) and data pulls, Oct 2026

Weather data by Open-Meteo.com (CC BY 4.0).